Lesson 2.

.gitignore
Token / API
Data sources -> Common Crawl, fineweb (edu)
Quality Dimension -> Relevance, Quality, Diversity
.gitignore bu github ga push qilinmaydigan files ro'yxati.
Avvalo free trail sharoitida github 1ta repo uchun 100 MB berishini hisobga olsak massiv datasni push qila olmaymiz. Shuningdek API keys ni ham push qilmaslik zarur. Boshqalar foyda olmasligi uchun.
API keylarni push qilib yuborish Machine learningda costi og'ir bo'lmasligi mumkin ammo LLMda bu million+ dollorlik xato bo'lishi hech gap mas.
Data sources uchun bir nechta mashhur saytlar. mas/ wd va. har biri

Oziga xos.

1 Common crawl bu yerda raw web datas juda kop lekin uni collect va cleaning qilish short shuningdek eng birinchi data sources lardan hisob lanadadi.

2. fineWeb SLM uchun cleaned va good quality datalar bor ular web text lar.

3. fineWeb-Edu bu yerda asosan educational SLM u-n educational datas.

4. The Pile bu yerda esa books, paper, articles, code, web datalar general ~~SLM~~ SLM u-n

5. Dolma bundagi datasets pretraining expriments u-n

6. The Stack v2 coding uch ~~data~~ data source.

7. Arxive bu yerda academic yani

scientific ~~SLM~~ SLM u-n research papers kop.
Lekin bularni orasida ham eng yaxshilari
fineWeb clean, high quality web texts for general SLM.
Educational purpose u-n fineWeb-Edu
Code u-n Stack v2 va ~~#~~ arXiv
Scientific SLM u-n.

Quality Dimension funda sifatga e'tibor qaratilib katta dataset yaxshi modelga sabab degani emas. datani clean, deduplication qilish bilan data yetarli darajada train yoki pretrain u-n yetarli bolmaydi. Talab qilinadigan strategiyalar bor.
1. Relevance yani maqsad qilingan model uchun mos data tayyorlash.

2. Quality yani text gramatik to'g'ri va ma'noni to'g'ri bo'lishi kerak aks holda model ham noto'g'ri o'rganib qo'yadi.
3. Diversity yani ma'lumot faqat bir xil bo'lmasligi va datani turli sources dan olishi kerak yani web-site dan ham, books, articles, papers, code, news and so on.
4. factual quality yani factlar va ma'lumotlar xato bo'lmasligi kerak
5. Safety / toxicity yani textda harmful text qismlar insonga tahdid qiladigan so'zlar, xavfli materials bo'lmasligi kerak.
6. Deduplication takroriy matnlar ortiqsa resource sarfiga olib keladi.
7. Sensitivity bunda private data yani informations. ni trainga berib yubormaslik kerak.

Data mixture yani faqatgina 1ta domainga asoslanilgan data b-n train qilsa yetarli emas. Unga yagin domain data b-n ham train zarur. data sources turli xil joydan olish kerak ex: Web data, education data, ② Scientic data, coding u-n data yani bular odatda general train yoki pretrain u-n. Data faqat bir ~~soha~~ soha haqida bo'lib qolmasligi kerak yagin soha datalari b-n ham train zarur. Data mixture model design u-n zarur.

License yani datadan qanday maqsadlar u-n foydalana olish yoki foydalana olmaslik haqida. Check list qilib olsak o'zimizga yaxshi.

① dataset owner kim?

② Nimalarga ruxsat berilgan?
③ Boshqalarga share qilib birgalikda ishlatsak bo'ladimi?
④ Bizness u-n foydalansam bo'ladimi
⑤ O'zgarish kirita olamanmi?
⑥ Qanday tiyiqlar bor.
~~Model~~da data licence haqida 1 ta yangi filedа ma'lumot berish k-k.

My LLM:
Programming + AI/ML Tutor

Help with Python, Machine Learning and AI concepts

1. Choose Domain

Programming + AI/ML

Purpose: Explain concepts and generate code.
Users: Students & beginners.

2. Find Data Sources

- Python Docs
- NumPy Docs
- Scikit-learn Docs
- PyTorch Docs

License: Apache 2.0, ODC-BY

3. Check Quality

- Relevant content
- Remove duplicates
- Filter bad data

4. Check License

NPset-python
Apache 2.0

SmolPyEdu
ODC-BY

- Commercial use
- Modify
- Distribution

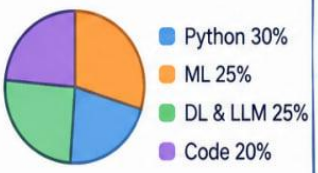
Attribution + license text

5. Track Provenance

Source name + license + date

- **AxiomicLabs/NPset-python**
  License: Apache 2.0
- **SmolPyEdu**
  License: ODC-BY

6. Design Mixture



7. Final Decision

USE
Focused domain, open licenses

Summary

- **Domain:** Programming + AI/ML
- Sources: Open & Legal
- Goal: Small, Useful LLM